



Mrinal Pandey

Chief Engineer

+917276002039

mrinal_61@rediffmail.com

Pune ,India

DOB -1983-07-10

Result-oriented techno -managerial leader with a proven track record in new product development ,vehicle integration and portfolio management within the automotive industry ,seeks a challenging leadership position .Expertise in driving innovation ,leading cross functional team and delivering exceptional products will contribute to orinizational success .Proven ability to manage complex project , optimize product design and implement cost-saving initiatives to maximize profitability and efficiency Passionate about contribution to the development of cutting edge automotive solutions that exceeds customer expectations and drive industry advancement .

Profile Summary

- A seasoned engineering professional with over 17 years of experience in the automotive industry specializing in New Product development ,vehicle integration and portfolio management for commercial vehicles specifically buses (Diesel ,Electric and Hybrid)
- Proven ability to lead cross-functional teams , manage complex projects and deliver innovative solutions with in cost and time constraints .
- Successfully led the emission migration of ILCV buses fromBS4 to BS6 standards , incorporating over 25 value enhancers ,exceeding expectations in a time and cost sensitive market .
- Conceptualized and demonstrated Tata Motors first double decker full electric bus with a portal axle ,showcasing expertise in electric vehicle technology and rapid proto typing.
- Possessing a strong understanding of market trends and customer needs , conducted QFDanalysis for the UAE market resulted in product development aligned with customer demands.
- Successfully implemented cost reduction initiatives ,including part standardization ,generating significant cost saving .
- Adept at managing all aspects of vehicle development from concept design and business case preparation through to digital validation , soft design release , prototype development ,testing and product launch .
- Proficient in utilizing various engineering tools and methodologies such as HOQ , DFMEA ,Risk analysis , 8-D and virtual validation techniques(CFD, CAE & NVH) to ensure product quality and safety .
- Proven ability to identify and mitigate technical risks ,developing and implementing effective mitigation plans throughout the vehicle development life cycle .
- Demonstrated leadership in driving organizational initiatives ,spearheading Bus VIG audits ,WCQ , Modularity, VAVE and Innovation initiatives .Showcasing a commitment to continuous improvement .
- Exceptional communicator ,and collaborator, effectively leading cross functional teams (Engineering , Product Line , Sourcing , Service ,Quality and Marketing) and providing comprehensive status to senior management .
- Strong technical aptitude with expertise in the areas such as vehicle dynamics , powertrain system and thermal management.
- Committed to mentoring and developing junior engineers ,fortering a collaborative and knowledge sharing envionrment .

Soft Skills

- Leadership
- Teamwork
- Communication
- Problem Solving
- Decesion Making
- Innovation
- Project Mnagment
- Analytical Skills
- Mentorship
- Goa Oriented

Technical Skills

- Vehicle Integration
- Product Development
- Digital Validation
- DFMEA
- MS Project
- PLM
- Product Costing

Core Competencies

- Automotive Engineering
- Vehicle Development
- Innovation and Design
- Cross Functional Leadership
- Strateic planning and Execution

Career Timelines

- Apr '2017- Present

DGM Vehicle Integration and Portfolio Management Buses (Diesel and Electric)
Tata Motors Limited ,India
- Apr '2013-Mar'17

Senior Manager , Ultra Buses (Domestic and IB IC buses)
Tata Motors Limited ,India
- Aug '2007-Mar'13

Manager Development , Ultra Trucks and Buses (Domestic and IB Market)
Tata Motors Limited ,India

Work Experience

Apr'2017- Present
DGM Vehicle Integration and Portfolio Management Buses (IC and Electric)
Tata Motors Limited , India, Pune

Key Result Area as a DGM /Vehicle Integration Commercial Vehicle

Certification

- PMP - Pursuing (Udemy)
- Google Certification Course in Project Management
- 3 Numbers Patent in DD EV Design
- Patent on modular design (2009)
- Paper Published in APM Russia 2007
- AIR -61 in GATE 2005

Education

M.Tech (Design)
IIT Bombay
Jul 2005- Jul2007

Executive MBA (Analytics)
IIM Nagpur ,Pune Campus Hybrid
Pursuing (2025-2027 Batch)

BE with Hon. (Mechanical)
BIT Durg
Aug 2001- Jul2005

Tata Motors · Full-time Tata Motors · Full-time Apr 2017 - Till Now · 5 yrs. 8 Months

Summary:

1. Delivered Emission migration of ILCV buses from BS4 to BS 6 with 25 + Value Enhancers
2. Defined new RE Midi Platform for IB product offering
3. Lead team for new product development and delivery .Lead QFD team for capturing input from UAE market and travelled Abroad .

Project Title: Ultra Buses - BS6 , RE Midi E4 IB bus , ILCV RDE &ESC compliance Buses 7T to 14t ,Double Decker EV Bus , 7m Electric Feeder Bus

Key Result Areas: Component design , Vehicle Integration, Project management .

1. Handled Program and Product Integration function for BS6 migration and Rear engine Bus project
2. Responsible for Initial concept design prepration, calculation ,Bussiness case study, resource estimation, budgeting , design freeze , DMU sign off, gateway plan ,benchmarking , proto plan ,target setting , specification finalization ,test plan ,design review and Status update to senior management
2. Doube Decker EV bus concept prepration and Demo vehicle build . Design Release .
2. Planned and executed parts standardization across variants leading to a cost-saving of 20 Cr INR .
3. Planned and executed cost-saving work shop to offset commodity price rise
5. Benchmarked R&D knowledge and published work instructions and guidelines for knowledge transfer.
6. Lead the team for achieving modularity BOM release
7. Lead the CFT team including Engineering , Product line ,Sourcing ,Service and Quality for program goal achievement

HSC

MP Board ,Raipur

Jun200

Jun200

Languages

Hindi

English

Hobbies

> Mentoring and Coaching

>Automotive Technology Enthusiast

> Travelling

Apr'2013- Mar'17

Senior Manager , Ultra Buses

Tata Motors Limited , India, Pune

Key Result Area as a Senior Manager /Vehicle Integration Ultra Buses

Tata Motors · Full-time Tata Motors · Full-time Apr 2013 - Mar 2017 · 5 yrs. 8 Months

Summary:

1. Handled design and development of Ultra BS4 Buses .
2. Handled vehicle packaging of FESLF bus from concept preparation to product delivery
3. Lead team for new product development and delivery

Project Title: Ultra Buses - BS4 , Multiplexing ,AMT

Key Result Areas: Component design , Vehicle Integration, Project management .

1. Defined requirements and interfaces to the environment. (Volume, weight, location, fasteners etc)
2. Led Vehicle package sign off activities for BS4 ,FESLF and AMT vehicles.
3. Generated decision matrix based on functional and commercial parameters for decision making.
4. Monitored Virtual Vehicle layout & finalize P&H (Proportions & Hard points).
5. Led DMU assessment for performance, manufacturability, and serviceability.
6. Spearheaded Geometric Integration activities & under-hood customer touch and fill points for Sign off.
7. Managed Variants & meet project specific commonization & standardization targets.
8. Defined, Monitored and Controlled the activities to ensure targeted deliveries are completed on time
9. Vehicle Design specs document generation and support for Homologation.
10. Interact with simulation teams (CFD, CAE & NVH) for virtual validation and design feedback.
11. Lead mule/VP/VP0/PP/SOP builds planning, coordination and issue resolution.
12. Monitor Vehicle Thermal Management activities for thermal and fire safety in vehicles.
13. Anticipate key technical risks and failure modes well in advance through DFMEAs and Risk Analysis to mitigate potential risks identified by defining mitigation plan in the integration domain
14. Lead resource planning, queries resolution, and workflow management within the team.
15. Collate and Publish monthly status of integration activities for program status assessment.
16. Benchmark relevant vehicles as per requirement and targets cascading
17. Lead Engine Support Systems (Air Intake, Exhaust, Park brake, Vacuum etc.) development.
18. Managing Vehicle DVPs and participating in vehicle build and validation activities.

Aug '2007-Mar'13

Manager Development , Ultra Trucks and Buses

Tata Motors Limited , India, Pune

Key Result Area as a Vehicle Integration ,Design and Development Manager / Ultra Trucks & Buses

Tata Motors · Full-time Tata Motors · Full-time Aug 2007 - Mar 2013 · 5 yrs. 8

▪ **Summary:**

1. Handled design and development of Exhaust system ,fuel system and frame for Ultra Range of new gen vehicles .
2. Handled vehicle packaging of Ultra range of buses from concept preparation to product delivery
3. Mentored and trained junior engineers on the software and project management skills

Project Title: New Gen Ultra Truck and Buses

Key Result Areas: Component design , Vehicle Integration, Project management .

1. Perform Layout Management and Packaging of components.
2. Execute Power train sizing, hard points sign off to conclude on powertrain architecture.
3. Generate interface sign off sheets for clearance issues and thermal issues.
4. Interacting with vendors for quotation & specification development.
5. Conducting benchmarking activities and gathering information for the product specs.
6. Handling the packaging issues and conducting concern resolution meetings.
7. Support and Coordination with Proto shop for vehicle build and testing.
8. Managed a team of 8 Engineers dedicated to Product Development